This Zenodo deposit contains a publicly available description of the Dataset:

"Single-cell RNAseq of human iPSC-derived wild type and PINK1 KO myeloid cells after lipopolysaccharide and interleukin-1 beta challenge".

Dataset Description:

We performed 10X Genomics single-cell RNAsequencing of human iSPC-derived monocytes and macrophages in vitro. Cells were treated with 500 ng/mL lipopolysaccharide (LPS) and 50 ng/mL interleukin-1 beta (IL1b) for 24 hours. This dataset contains raw FASTQ files from myeloid cells, sorted into 4 groups namely monocytes (Mono) and Macrophages (Mac) non-stimulated (NS) and LPS+IL1b-stimulated cells. Sequencing was performed using NovaSeq 6000 S4 PE 100bp. Reads were processed using the 10X Genomics Cell Ranger Single Cell 2.0.0 pipeline. FASTQs generated from sequencing output were aligned to the human GRCh38 reference genome using STAR algorithm 2.7.3a.

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Dataset Name: desjardins-ipsc-sc-rnaseq-myeloid-pink1, vv0.1

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ASAP Lab:

ASAP Project: Myeloid PINK1 represses mtDNA release and immune signaling that impacts neuronal pathology in patient-derived idiopathic PD models

Project Description: Parkinson's disease (PD) is a neurodegenerative disorder marked by the development of cardinal motor deficits preceded by a protracted prodromal period of non-motor symptoms often involving the gastrointestinal (GI) tract. There is an emerging consensus that both the peripheral immune system and local neuroinflammation play key roles in the etiology of PD. We previously demonstrated a critical function for the Parkinson's related proteins PINK1 and Parkin as repressors of the innate to adaptive immune response in cultured cells and mouse models of infection. However, it remained unclear whether these processes were conserved in patient-derived models, and precisely how immune signaling may ultimately drive the death of dopaminergic neurons. Here we show that GI infection of PINK1 knockout (KO) mice triggered acute neurodegeneration which was evident early in the enteric nervous system. Treating wild type enteric or dopaminergic neurons with conditioned medium from immune-stimulated PINK1 KO macrophages was sufficient to promote neuronal disruption in both mouse and human neurons in vitro. Within immune-activated macrophages, we reveal that loss of PINK1 led to an enhanced release of mitochondrial DNA (mtDNA) within mitochondrial derived vesicles, leading to the activation of cGAS/STING pathways. These changes were seen in both mouse/human in vitro models and in PD patient-derived primary macrophages. Notably, pharmacological modulation using a PINK1 activator with high therapeutic potential attenuated pro-inflammatory profiles elicited by the mtDNA-dependent STING/NF-Î°B pathway in idiopathic patient-derived macrophages. Ultimately, our study lays the foundation for understanding PINK1-related peripheral macrophage mechanisms in idiopathic PD and provides a target for further development to treat the disease at early stages.

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